Math 261-001
Quiz 7, October 23

Name	CSU ID #

Choose and answer **exactly two** of the four problems. If you attempt more than two, only the first two will be graded. Write your final answers in the boxes provided. Only answers in the boxes will be graded. You may use the surrounding space for scratch work. Each problem is worth the same number of points.

1. For real constants a, b, let

$$\vec{F}_{a,b}(x, y, z) = (2xy + az^2,\ bx^2 + z,\ 2xz + y).$$

Find all pairs (a, b) for which $\vec{F}_{a,b}$ is conservative. For those values, find a potential function f.

(a, b)

Potential $f(x, y, z)$

2. Let $\vec{F}(x, y) = (y^3 - 2y,\ -4x + 3y^2)$, and let C be the counterclockwise boundary of the rectangle $0 \le x \le 2$ and $0 \le y \le 1$. Find the circulation $\oint_C \vec{F} \cdot \vec{T}\, ds$ and the outward flux $\oint_C \vec{F} \cdot \vec{N}\, ds$, where $\vec{T}$ is the unit tangent in the direction of traversal and $\vec{N}$ is the outward unit normal.

Circulation

Outward flux

3. Let

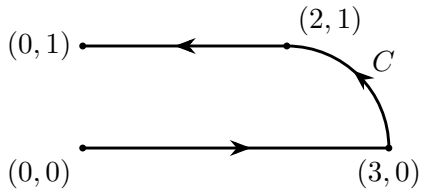
$$f(x,y,z) = x^2 + y^2 + z^2, \qquad \vec{G}(x,y,z) = (z^2, x^2, y^2).$$

Compute $\nabla \cdot \big(f(\nabla \times \vec{G})\big)$.

4. Use Green's theorem to evaluate $\displaystyle\int_C \vec{F} \cdot d\vec{r}$, where

$$\vec{F}(x,y) = \big(ye^{xy} - y,\ xe^{xy} + x + 1\big).$$

The path C follows, in order, the straight segment $(0,0) \to (3,0)$, the counterclockwise quarter-circle $(x-2)^2 + y^2 = 1$ from $(3,0)$ to $(2,1)$, and the straight segment $(2,1) \to (0,1)$.



Reminder: $\displaystyle\iint_R 1\, dA = \text{Area}(R)$.